

Stewart C. Jamieson

PHD CANDIDATE · AUTONOMOUS ROBOTS & MACHINE LEARNING

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A roboticist working to develop autonomous systems that co-operate with humans in uncertain, unstructured, and dynamic environments to accomplish their objectives safely and efficiently.

Research Interests & Skills

Artificial Intelligence	Online & Active Learning, Unsupervised Learning, Robust Vision, Spatiotemporal Topic Modelling, AI Ethics
Robotics	Human-Robot Collaboration, Autonomous Exploration, Uncertainty Quantification, Informative Path Planning
Programming	Python, C++20, PyTorch, Keras, ROS, OpenCV, Pandas, Eigen, MATLAB, Mathematica, Bash, Java, Android

Education

Massachusetts Institute of Technology & Woods Hole Oceanographic Institution

Cambridge, MA, USA

PH.D. IN APPLIED OCEAN SCIENCE AND ENGINEERING — MARINE ROBOTICS (5.0/5.0 CGPA)

June 2020 - Present

- Thesis: *Human-Guided Multi-Robot Scientific Exploration in Bandwidth-Limited Environments*
- Developing multi-robot systems to collaboratively explore unfamiliar environments with limited human guidance
- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)

S.M. IN AERONAUTICS AND ASTRONAUTICS — AUTONOMOUS SYSTEMS (5.0/5.0 CGPA)

June 2018 - May 2020

- Thesis: *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*
- Co-supervised by Dr. Yogesh Girdhar (WHOI) and Prof. Jonathan P. How (MIT)
- Relevant Coursework: Cognitive Robotics, Visual Navigation for Autonomous Vehicles, Bayesian Modelling & Inference

University of Toronto

Toronto, ON, Canada

B.A.SC. IN ENGINEERING SCIENCE WITH HONOURS — ROBOTICS MAJOR (3.83/4.0 CGPA)

Sept. 2013 - Apr. 2018

- Thesis: *Deep Learning for Robust Vision in Realtime Autonomous Driving*, supervised by Prof. Angela Schoellig
- Applied state-of-the-art uncertainty quantification techniques to achieve safer autonomous driving performance

Work & Research Experience

WHOI's Autonomous Robotics and Perception Laboratory (WARPLab)

Woods Hole, MA, USA

GRADUATE RESEARCH ASSISTANT

June 2018 - Present

- Developing autonomous exploration algorithms for multi-robot teams exploring the deep ocean and coral reefs
- Publications focus on enabling robot co-operation with humans over slow, bandwidth-limited communication channels
- Assisting with the deployment of these novel algorithms into WHOI's world-class deep sea exploration vehicles

aUToronto Self-Driving Car Team (SAE/GM AutoDrive Challenge)

Toronto, ON, Canada

SOFTWARE TEAM LEAD

June 2017 - June 2018

- At the end of my term, aUToronto won **1st place in the SAE/GM AutoDrive Challenge (Year 1)**
- Led a subteam of 12 graduate and undergraduate students working to develop an autonomous Chevrolet Bolt
- My team created the overall system software architecture, sensor drivers, vehicle control interface, and software services

Zebra Technologies Inc.

Mississauga, ON, Canada

SOFTWARE ENGINEERING INTERN, ENGINEERING PRODUCT INNOVATION TEAM

May 2016 - Aug. 2017

- Co-designed and co-developed the software of the first Zebra SmartSight™ prototype, a robot designed for retail operations
- Helped to research and present business applications for robotics, machine learning, and neural networks
- 16 months of C++14 development experience including networking, databases, and high performance computing

Wattpad Inc.

Toronto, ON, Canada

ANDROID SOFTWARE DEVELOPER, READER ACQUISITION TEAM

May 2015 - Sept. 2015

- Wattpad is a worldwide storytelling platform with a community of over 80 million users
- Implemented features designed to attract new users; also implemented A/B tests to validate each features' success

- Searched for, reported, and fixed software bugs in the Android mobile application with over 15 million users
- Designed and implemented a virtual doorman to greet company visitors and notify staff of their arrival

Publications

PEER-REVIEWED CONFERENCE PAPERS

Jamieson, S., How, J. P., & Girdhar, Y. (2023). *DeepSeaColor: Realtime Adaptive Color Correction for Autonomous Underwater Vehicles via Deep Learning Methods*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Girdhar, Y., McGuire, N., Cai, L., **Jamieson, S.**, McCammon, S., San Soucie, J. E., Todd, J. E., Claus, B., & Mooney, T. A. (2023). *CUREE: A Curious Underwater Robot for Ecosystem Exploration*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Jamieson, S., Fathian, K., Khosoussi, K., How, J. P., & Girdhar, Y. (2021). *Multi-Robot Distributed Semantic Mapping in Unfamiliar Environments through Online Matching of Learned Representations*. In 2021 IEEE International Conference on Robotics and Automation (ICRA). Xi'an, China.

Jamieson, S., How, J. P., & Girdhar, Y. (2020). *Active Reward Learning for Co-Robotic Vision Based Exploration in Bandwidth Limited Environments*. In 2020 IEEE International Conference on Robotics and Automation (ICRA). Paris, France. **Won Best Paper Award in Service Robotics.**

Girdhar, Y., Cai, L., **Jamieson, S.**, McGuire, N., Flaspohler, G., Suman, S., & Claus, B. (2019). *Enabling Co-Robotic Scientific Exploration of Unknown Environments over a Low Bandwidth Communication Channel*. In 2019 IEEE International Conference on Robotics and Automation (ICRA). Montréal, Canada.

WORKSHOP PAPERS

Jamieson, S., Todd, J. E., How, J. P., & Girdhar, Y. (2021). *Communicating Efficiently to Enable Human-Multi-Robot Collaboration in Space Exploration*. In "SpaceCHI: Human-Computer Interaction for Space Exploration" Workshop at CHI 2021. Yokohama, Japan.

Beaulieu, S., Alexander, H., **Jamieson, S.**, Longworth, B., McLean, C., Soenen, K., York, A., Krinos, A., Cai, L., Govostes, R. and & Hernandez, C. (2020). *Building a data science curriculum and community for ocean scientists, engineers, and students using The Carpentries model*. In AGU Fall Meeting 2020.

Jamieson, S. (2019). *The Pervasiveness of Deep Learning in Robotics Research Does Not Impede Scientific Insights into Robotics Problems*. In "Debates on the Future of Robotics Research" Workshop at ICRA 2019. Montréal, Canada.

THESES

Jamieson, S. (2020). *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*. Master's Thesis, Massachusetts Institute of Technology & Woods Hole Oceanographic Institution.

Jamieson, S. (2018). *Deep Learning for Robust Vision in Realtime Autonomous Driving*. B.A.Sc. Thesis, University of Toronto.

Honors & Awards

INTERNATIONAL

- 2020 **Best Paper Award in Service Robotics (out of 1483 Accepted Papers)**, ICRA 2020
 2018 **1st Place Team**, SAE/GM AutoDrive Challenge

Paris, France
Yuma, AZ, USA

DOMESTIC

- 2018 **Dean's Honour List**, University of Toronto
 2014-16 **Dean's Honour List (x3)**, University of Toronto
 2013 **Governor General's Bronze Medal for Academic Excellence**, High School Graduation
 2013 **Regional Champion**, ECOO Programming Competition
 2010-13 **School Champion (x4)**, Waterloo CEMC Math Contest

Toronto, ON, Canada
Toronto, ON, Canada
Burlington, ON, Canada
Halton, ON, Canada
Burlington, ON, Canada

Presentations

Videos and other materials used in some of the following presentations are available at www.stewartjamieson.com

ROBOTS: How new technologies are advancing ocean research, exploration, and science-based solutions

[Online](#)

PRESENTER, WHOI OCEAN ENCOUNTERS (SEASON 5 EPISODE 4)

May 2023

Multi-Robot Reef Monitoring, Mapping, & Human-Guided Exploration

[Online](#)

INVITED TALK, NCCOS SFM/AI WORKING GROUP

March 2023

Rapid, Comprehensive Coral Reef Monitoring with AUVs

[Bremen, Germany](#)

INVITED TALK, 15TH INTERNATIONAL CORAL REEF SYMPOSIUM

July 2022

Communicating Efficiently to Enable Human-Multi-Robot Collaboration

[Online](#)

POSTER PRESENTATION, SPACECHI WORKSHOP @ CHI 2021

May 2021

Human-Robot Cooperation for Exploring Bandwidth-Limited Environments

[Woods Hole, MA, USA](#)

INVITED TALK, WHOI AOPE DEPARTMENT SEMINAR SERIES

July 2020

Active Reward Learning for Co-Robotic Exploration in Bandwidth-Limited Environments

[Cambridge, MA, USA](#)

INVITED PRESENTER & PANELIST, ICRAxMIT

June 2020

Deep Learning Does Not Impede Scientific Insights into Robotics Problems

[Montréal, QC, Canada](#)

INVITED LIGHTNING TALK, DEBATES ON THE FUTURE OF ROBOTICS RESEARCH, ICRA 2019

May 2019

16.412 Lecture: Multi-Robot Adaptive Sampling

[Cambridge, MA, USA](#)

CO-LECTURER, MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Apr. 2019

An Introduction to Neural Networks and Machine Learning

[Mississauga, ON, Canada](#)

LEAD PRESENTER FOR ZEBRA TECHNOLOGIES INC. "LUNCH & LEARN" (200+ ATTENDEES)

Jan. 2017

Should Robots Have Rights?

[Toronto, ON, Canada](#)

CO-PRESENTER IN DEBATE AT THE UNIVERSITY OF TORONTO

Dec. 2015

A Customized Graphical Checklist for Efficient Ambulance Inventory

[Toronto, ON, Canada](#)

CO-PRESENTER IN "PRAXIS II SHOWCASE" AT THE UNIVERSITY OF TORONTO

Apr. 2014

Teaching

Massachusetts Institute of Technology

[Worldwide](#)

TEACHING ASSISTANT

Aug. 2020 - Dec. 2020

- Fall 2020: 16.485 Visual Navigation for Autonomous Vehicles (Prof. Luca Carlone)

The Carpentries

[Worldwide](#)

CERTIFIED SOFTWARE CARPENTRIES INSTRUCTOR

Oct. 2019 - Present

- June 2021: Co-Instructor for WHOI Student Python Workshop Series 2021
- Oct. 2020: Helper for WHOI Data Carpentry Workshop Fall 2020
- July 2020: Co-Instructor for WHOI Student Python Workshop Series 2020
- Oct. 2019: Helper for WHOI Software Carpentry Workshop Fall 2019

Academic Service

- Organizer of:
 - AI for Earth & Space Science Workshop @ ICLR 2022
- Reviewed Journal Submissions for:
 - Robotics and Automation Letters (RA-L)
 - Journal of Aerospace Information Systems (JAIS)
- Reviewed Conference Submissions for:
 - International Conference on Learning Representations (ICLR)
 - Conference on Neural Information Processing Systems (NeurIPS)
 - International Conference on Machine Learning (ICML)
 - International Conference on Robotics and Automation (ICRA)
 - International Conference on Intelligent Robots and Systems (IROS)

Professional Service

MIT-WHOI Joint Program

ELECTED AT-LARGE REPRESENTATIVE

Woods Hole, MA, USA

Oct. 2020 - Oct. 2021

- Co-hosted monthly online student events to maintain social cohesion during worldwide pandemic
- Developed and published the MIT-WHOI Joint Program online photoboard

Zebra Technologies Inc.

EDITOR, EMC INNOVATION NEWSLETTER

Mississauga, ON, Canada

May 2016 - Aug. 2017

- Edited bi-monthly department newsletter and distributed it to over 1700 engineers
- Commissioned, reviewed, and published articles about recent trends and innovations in electronics, robotics, etc.

Graduate Student Member

IEEE (INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS)

Worldwide

Nov. 2013 - Present

Personal Interests

Corpus Christi Jazz Horns and Concert Band

SAXOPHONIST

Burlington, ON, Canada

Sept. 2009 - June 2013

- Performed in the Atlantic Music Festival (2013), Toronto Music Festival (2012)